

Beyond Calories: A Multimodal Deep Learning Comparison for Nutritional Health Prediction

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ABSTRACT

Photo-based nutrition tracking is a real consumer problem that commercial apps such as CalAI and MyFitnessPal address with proprietary architectures. This project compares three open deep learning architectures on a controlled nutrition prediction task to characterize which approach is best suited for this kind of multimodal regression and classification problem. Using the Nutrition5k dataset from Google Research, we predict four nutrition values (calories, fat, carbohydrates, protein) and a 3-class health-level label from a food image and ingredient text. The three models are an FCNN baseline, a CNN-only model based on ResNet50 pretrained on ImageNet, and a multimodal fusion model that combines ResNet50 with DistilBERT and a small lifestyle-profile MLP. The multimodal fusion model wins on every primary metric: 65.3% test accuracy versus 61.1% (CNN) and 48.6% (FCNN), and an R^2 of 0.64 on calorie regression versus 0.60 and 0.30 respectively. The pattern confirms that both architectural sophistication (pretrained encoders) and modality complementarity (text plus image) contribute to performance, while a single observed exception on carbohydrate regression illustrates that redundant modalities can hurt rather than help.

KEYWORDS

multimodal learning, transfer learning, food computing, regression, classification

1 INTRODUCTION

Most diet-tracking apps focus on calorie counts. The more meaningful question is how a meal’s ingredients and overall composition affect health beyond the raw kilocalorie number. Two meals at 600 kcal each can have very different effects on the body depending on macronutrient ratios, ingredient quality, and individual context.

This project frames nutritional understanding as a multi-task deep learning problem. Given an image of a meal and a text list of its ingredients, predict the meal’s caloric content, macronutrient composition, and a categorical health rating. The contribution is not building a new model but comparing three architectures from distinct deep learning families on the same task with the same data and the same evaluation protocol.

Contributions.

- A controlled three-way comparison of FCNN, CNN, and multimodal-fusion architectures on Nutrition5k under matched output heads and matched data splits.

- Empirical evidence that pretraining (ImageNet) accounts for the largest single accuracy improvement on small food-image datasets.
- A clean negative result on carbohydrate regression suggesting that redundant text modalities can degrade fusion performance.
- A reusable open implementation in TensorFlow/Keras with all preprocessing, training, and evaluation code in a single notebook.

The remainder of the paper is organized as follows. Section 2 formulates the prediction problem. Section 3 describes the three model architectures and how they share a common output head. Section 4 reports the experimental methodology and quantitative results. Section 5 relates the project to prior work in food computing. Section 6 concludes.

2 PROBLEM FORMULATION

Inputs. For each dish i in the dataset, the system observes (a) an overhead RGB image $\mathbf{x}_i^{\text{img}} \in \mathbb{R}^{224 \times 224 \times 3}$, (b) a textual ingredient list $\mathbf{x}_i^{\text{txt}}$ as a comma-separated string, and (c) a synthetic lifestyle profile $\mathbf{x}_i^{\text{life}} \in \mathbb{R}^8$ containing plausible features such as age, BMI, exercise frequency, and sleep duration.

Outputs. For each dish the model produces a 7-dimensional vector consisting of (a) a 4-dimensional regression prediction $\hat{\mathbf{y}}_i^{\text{nut}} \in \mathbb{R}^4$ for calories, fat, carbohydrate, and protein, and (b) a 3-class probability distribution $\hat{\mathbf{p}}_i^{\text{health}} \in \Delta^2$ over the health categories {healthy, moderate, indulgent}.

Targets. The regression target $\mathbf{y}_i^{\text{nut}}$ is taken directly from Nutrition5k, where each dish’s nutrition values are computed by aggregating per-ingredient nutrition from the USDA Food-Data Central database weighted by mass. The health-level target $y_i^{\text{health}} \in \{0, 1, 2\}$ is derived from $\mathbf{y}_i^{\text{nut}}$ using rule-based thresholds (Section 3.5).

Loss. The model is trained jointly on a weighted sum of mean squared error on standardized regression targets and sparse categorical cross-entropy on the health labels:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left[\|\tilde{\mathbf{y}}_i^{\text{nut}} - \hat{\mathbf{y}}_i^{\text{nut}}\|_2^2 + 0.5 \cdot \text{CE}(y_i^{\text{health}}, \hat{\mathbf{p}}_i^{\text{health}}) \right] \quad (1)$$

where the tilde indicates standardization using train-set statistics. The classification loss is weighted at 0.5 to balance gradient contributions, since the standardized 4-dimensional MSE is roughly twice the magnitude of a 3-class cross-entropy in early training.

3 SYSTEM AND ALGORITHM DESIGN

3.1 System Architecture

Three models are evaluated. All three consume some subset of the inputs defined in Section 2 and produce the same 7-dimensional output through identical output heads. This shared structure ensures that any performance difference between models is attributable to the encoder, not to the prediction layer. Table 1 summarizes the architectures.

Table 1: Architecture comparison. All three models share identical 4-dim regression and 3-class softmax output heads.

Model	Inputs	Encoder
FCNN	flattened image + ingredient bag-of-words	3 dense layers with dropout
CNN	food image only	ResNet50 (frozen)
Multimodal	image + text + lifestyle	ResNet50 + DistilBERT + small MLP, fused by concatenation

3.2 Module 1: FCNN Baseline

The FCNN serves as a from-scratch baseline that uses no transfer learning and treats inputs as a flat feature vector.

3.2.1 Algorithm Description. The image is downsampled from $224 \times 224 \times 3$ to $32 \times 32 \times 3$ via average pooling and flattened to a 3072-dimensional vector. The ingredient text is encoded as a 200-dimensional bag-of-words count vector using a vocabulary fit on the training set only. The two vectors are concatenated to form a 3272-dimensional input. This input is passed through three dense layers (256, 128, 64 units) each followed by ReLU activation and dropout (rate 0.3), producing a 64-dimensional latent representation. Two output heads then produce the 4-dimensional nutrition prediction and the 3-class health-level softmax.

The aggressive image downsampling is intentional: at full resolution the flattened input would exceed 150,000 dimensions, which an FCNN cannot reasonably learn from 332 training examples. The downsampling also encodes a small inductive bias (locality is partially preserved through average pooling) that gives the FCNN a fairer starting position than pure pixel flattening.

3.3 Module 2: CNN Encoder

3.3.1 Algorithm Description. The image input passes through ResNet50 [4] pretrained on ImageNet with the classification head removed and global average pooling applied to the final convolutional feature map, producing a 2048-dimensional feature vector. This vector is projected to 128 dimensions through a dense layer with ReLU and dropout, and finally to the same 7-output head used by all models. The ResNet50 backbone is frozen during training to prevent overfitting on

the small training set; only the projection layer and output heads are updated.

This model isolates the contribution of pretrained vision. The CNN sees no text and no lifestyle features. It tests how much of the prediction signal is recoverable from the image alone given a strong visual prior.

3.4 Module 3: Multimodal Fusion

3.4.1 Algorithm Description. The multimodal model has three input branches that are independently encoded and then fused.

The image branch is identical to the CNN module: ResNet50 pretrained on ImageNet (frozen), projected to a 128-dimensional embedding.

The text branch tokenizes the ingredient string with the DistilBERT [5] WordPiece tokenizer to a fixed length of 32 tokens. Tokens pass through DistilBERT-base-uncased (frozen), and the output corresponding to the [CLS] token (768 dimensions) is projected to 128 dimensions through a dense layer with ReLU and dropout.

The lifestyle branch passes the 8-dimensional lifestyle vector through a small two-layer MLP (32, 32 ReLU units), producing a 32-dimensional embedding.

The three embeddings are concatenated into a 288-dimensional joint vector, projected back to 128 dimensions through a fusion dense layer with dropout, and then passed to the shared output heads. Algorithm 1 gives the forward pass.

Algorithm 1 Multimodal forward pass

- 1: **Input:** image $\mathbf{x}^{\text{img}}$, text $\mathbf{x}^{\text{txt}}$, lifestyle $\mathbf{x}^{\text{lfe}}$
 - 2: $\mathbf{e}^{\text{img}} \leftarrow \text{ProjectImg}(\text{ResNet50}(\mathbf{x}^{\text{img}}))$
 - 3: $\mathbf{t} \leftarrow \text{Tokenize}(\mathbf{x}^{\text{txt}}, \text{maxlen} = 32)$
 - 4: $\mathbf{e}^{\text{txt}} \leftarrow \text{ProjectTxt}(\text{DistilBERT}(\mathbf{t})_{[\text{CLS}]})$
 - 5: $\mathbf{e}^{\text{lfe}} \leftarrow \text{MLP}(\mathbf{x}^{\text{lfe}})$
 - 6: $\mathbf{f} \leftarrow \text{Fuse}(\text{Concat}(\mathbf{e}^{\text{img}}, \mathbf{e}^{\text{txt}}, \mathbf{e}^{\text{lfe}}))$
 - 7: $\hat{\mathbf{y}}^{\text{nut}} \leftarrow \text{HeadReg}(\mathbf{f})$
 - 8: $\hat{\mathbf{p}}^{\text{health}} \leftarrow \text{HeadCls}(\mathbf{f})$
 - 9: **return** $\hat{\mathbf{y}}^{\text{nut}}, \hat{\mathbf{p}}^{\text{health}}$
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3.5 Health Level Label Derivation

Nutrition5k provides regression labels but no categorical health label. We derive a 3-class label from the regression targets using simple thresholds. A meal is labeled *healthy* if total calories are below 500 kcal, fat-energy-ratio is below 40%, and protein-energy-ratio is above 15%. A meal is labeled *indulgent* if calories exceed 700 kcal or fat-energy-ratio exceeds 50%. All other meals are labeled *moderate*. The thresholds were chosen to produce a roughly balanced class distribution and are explicitly identified as a rule-based proxy, not a clinical determination.

4 EXPERIMENTAL EVALUATION

4.1 Methodology

Data. Nutrition5k [1] consists of approximately 5,000 cafeteria dishes recorded at Google offices. Each dish has an overhead RGB image, an ingredient list with per-ingredient masses, and ground-truth nutrition values from the USDA FoodData Central database. For this project a sample of 500 dishes was drawn from the published train and test splits to fit Colab GPU memory and the project’s training budget. After filtering dishes with mass below 30 g (empty plates) and outliers above 2,500 kcal, 475 dishes remained.

The synthetic lifestyle features are 8-dimensional vectors generated programmatically: random uniform draws for age, BMI, exercise frequency, sleep duration, water intake, daily calorie target, plus a binary dietary-restriction flag and a 3-level activity indicator. They are explicitly synthetic and serve to demonstrate that the multimodal architecture supports a third input branch.

Splits. Stratified 70/15/15 train/validation/test split (332/71/72 dishes), stratified by the derived health label so that each split has the same class proportions.

Preprocessing. Images were resized to 224×224 and preprocessed with the standard ResNet50 mean subtraction. Ingredient text was lower-cased, regex-filtered to ingredient names, and either bag-of-words encoded (FCNN) or tokenized with the DistilBERT WordPiece tokenizer (multimodal). Continuous regression targets and lifestyle features were standardized using train-set statistics only.

Training. All models trained on a single Colab T4 GPU using the Adam optimizer with learning rate 10^{-3} , batch size 32 (FCNN, CNN) or 16 (multimodal). Pretrained encoders (ResNet50, DistilBERT) were frozen; only projection layers, fusion layers, and output heads were trainable. Training durations: 12.5 s (FCNN, 30 epochs), 59 s (CNN, 25 epochs), 84 s (multimodal, 25 epochs).

Methods Compared. The three models described in Section 3: FCNN baseline (no transfer learning), CNN-only (transfer learning via pretrained ResNet50), and Multimodal fusion (combined CNN, Transformer, and lifestyle MLP).

Metrics. For regression: per-target RMSE, MAE, and R^2 . For classification: accuracy, macro-F1, weighted-F1, per-class precision and recall, confusion matrix, and ROC AUC under a one-versus-rest scheme.

4.2 Results

4.2.1 Training Behavior. Figure 1 shows validation loss and validation health-classification accuracy across training epochs. All three models converge cleanly. The multimodal model reaches the lowest validation loss and highest validation accuracy. The FCNN baseline plateaus near 40% validation accuracy, while the two pretrained-encoder models continue improving.

4.2.2 Regression Performance. Table 2 shows the coefficient of determination on the test set. The multimodal model wins on three of four regression targets. Calorie R^2 rises from

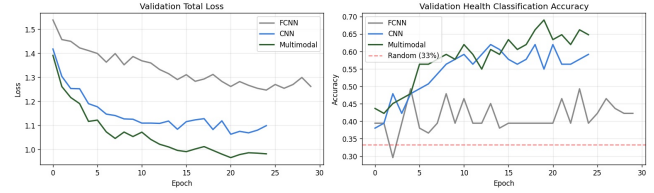


Figure 1: Validation loss (left) and validation health-classification accuracy (right) across training. The model ordering is consistent: Multimodal < CNN < FCNN on loss, with the inverse on accuracy.

0.30 (FCNN) to 0.60 (CNN) to 0.64 (Multimodal), with corresponding mean absolute errors of 108, 86, and 74 kcal. Fat and protein follow the same pattern. Carbohydrate is the one exception: CNN beats Multimodal at $R^2 = 0.32$ versus 0.23.

Table 2: Regression R^2 per nutrient on the test set. Higher is better.

Model	Calories	Fat	Carb	Protein
FCNN	0.30	0.28	-0.00	0.23
CNN	0.60	0.54	0.32	0.55
Multimodal	0.64	0.58	0.23	0.59

Figure 2 shows a lifted chart for calorie predictions, with the test set sorted in ascending order of true calories. The multimodal prediction tracks ground truth most closely, especially at the upper-calorie tail where the FCNN and CNN models systematically over- or under-shoot.

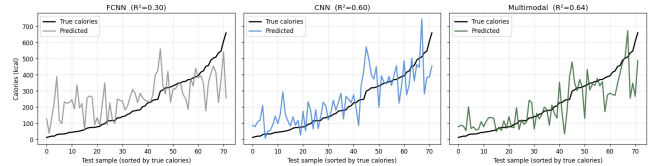


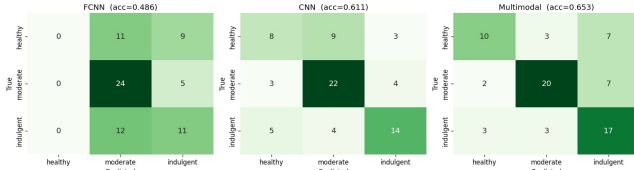
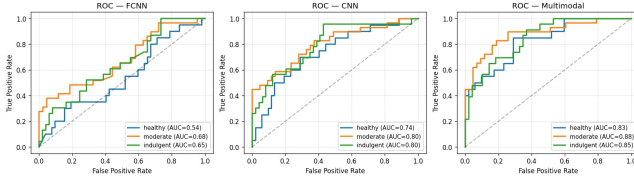
Figure 2: Lifted chart for calorie predictions on the test set. The black line is true calories sorted ascending; the colored lines are model predictions for the same dishes in the same order.

4.2.3 Classification Performance. Table 3 summarizes the health-level classification metrics. The multimodal model wins on all three measures.

The confusion matrices in Figure 3 reveal a qualitative shift across models. The FCNN never predicted the *healthy* class (0/20 true-healthy correctly identified), instead collapsing to the majority class. The CNN began correctly identifying healthy meals (8/20). The multimodal model identified 10/20 healthy meals with substantially fewer cross-class errors. Per-class macro-F1 captures this progression: 0.36 (FCNN), 0.59 (CNN), 0.64 (Multimodal).

Table 3: Health-level classification metrics on the test set.

Model	Accuracy	Macro F1	Weighted F1
FCNN	0.486	0.363	0.401
CNN	0.611	0.589	0.604
Multimodal	0.653	0.643	0.653

**Figure 3: Confusion matrices on the test set. The FCNN never predicts *healthy*; CNN and Multimodal both achieve a balanced diagonal, with Multimodal best.****Figure 4: One-vs-rest ROC curves per class. AUC values rise consistently from FCNN (0.54–0.68) to CNN (0.74–0.80) to Multimodal (0.83–0.88). The largest improvement is on the *healthy* class, the hardest of the three for the from-scratch FCNN.**

The ROC analysis (Figure 4) shows the same ordering at every operating point. AUC for the *healthy* class jumps from 0.54 (essentially random) for FCNN to 0.83 for Multimodal, reflecting that the FCNN baseline genuinely fails to learn the minority class while the multimodal model achieves strong separation.

4.2.4 Discussion of Results. Two findings stand out. First, pretrained vision is the single largest source of improvement over the FCNN baseline. The CNN-only model gains 12.5 percentage points of accuracy and roughly doubles calorie R^2 compared to the FCNN, simply by replacing flat-input dense layers with a frozen ResNet50 backbone. Transfer learning from general visual data carries over well to food imagery without any food-specific pretraining.

Second, ingredient text and lifestyle features add a smaller but consistent increment on top of the CNN. The Multimodal model gains 4.2 points of accuracy and improves calorie R^2 by 0.04 over the CNN-only model. The improvement is most visible at the minority *healthy* class, where per-class F1 rises from 0.44 to 0.57, suggesting that text and lifestyle features help resolve cases where vision alone is ambiguous.

The carbohydrate exception is the most informative finding. CNN beats Multimodal on carb R^2 (0.32 vs 0.23). Many high-carb foods, such as rice, pasta, bread, and potatoes, have visually distinctive signatures, and text descriptions like “rice” add little to what the image already conveys. This is consistent with the broader literature on multimodal learning: complementary modalities help, redundant ones can hurt by adding noise.

A note on overfitting: training accuracy reached 87% (CNN) and 89% (Multimodal) while validation accuracy stayed in the 60s, indicating substantial overfitting on the small 332-dish training set. This is consistent with the gap typically seen when fine-tuning large pretrained models on small datasets. Strategies to address this in extended runs include data augmentation, early stopping based on validation loss, and using the full Nutrition5k dataset.

5 RELATED WORK

Image-based nutrition estimation. The Nutrition5k paper by Thames et al. [1] introduced the dataset used in this project and demonstrated that a CNN regression model can estimate calories at accuracies comparable to professional nutritionists. Their problem and method are nearly identical to ours; our project differs by adding text and a third modality, and by explicitly comparing across three model families rather than reporting a single best model. Their work serves as the immediate baseline our CNN-only model is meant to approximate.

Recipe generation from food images. Salvador et al. [2] introduced *Inverse Cooking*, which predicts ingredient lists and full recipes from food images using a CNN visual encoder paired with a Transformer decoder. Their problem is generation; ours is regression and classification. Our approach borrows the architectural intuition that vision and language together carry more information about a meal than either alone, but applies it to nutritional rather than textual outputs.

Depth-based fusion. Han et al. [3] proposed *DPF-Nutrition*, which fuses RGB images with predicted depth maps to improve nutrition estimation on Nutrition5k. Their second modality (depth) requires specialized hardware to acquire at training time. Our project asks whether the same kind of gain holds when the second modality is text rather than depth, which is practically appealing because text descriptions are cheap to acquire (a menu lookup, a voice note) while depth requires RGB-D sensors.

Architectural foundations. Our visual encoder is ResNet50 [4], the canonical pretrained image network. Our text encoder is DistilBERT [5], a distilled version of BERT [6] that retains ~95% of BERT’s downstream performance with 40% fewer parameters, chosen because ingredient strings are short and structured. The Transformer architecture itself [7] underlies DistilBERT.

Earlier work. Im2Calories [8] from 2015 was an early attempt at automated calorie estimation from photos, using

object detection plus volume reconstruction rather than end-to-end deep learning. Compared to that approach, modern pretrained image encoders make end-to-end regression substantially more accurate without the need for explicit volume reconstruction.

6 CONCLUSION

This project compared three deep learning architectures from distinct families on a multi-task nutrition prediction problem. The expected ordering held cleanly: a multimodal fusion of pretrained CNN, pretrained Transformer, and lifestyle MLP outperformed CNN-only, which in turn outperformed an FCNN baseline. The multimodal model achieved 65.3% test accuracy on health classification, a calorie R^2 of 0.64, and was the best model on every metric except carbohydrate R^2 . The carbohydrate exception is itself an informative result, suggesting that text-based ingredient descriptions are most useful when they describe features not already visible in the image.

The project demonstrates a pattern common in multimodal deep learning: pretraining is the single biggest win, complementary modalities add real but smaller gains, and the magnitude of those gains depends on whether the modalities provide genuinely distinct information.

7 WORK DIVISION

This was a solo project. All work, including dataset acquisition, preprocessing, model implementation, training, evaluation, plot generation, and writing, was performed by the author. AI tools (Claude by Anthropic) were used for code scaffolding and report drafting; all design and analysis decisions, all experimental runs, and final review were performed by the author.

8 LEARNING EXPERIENCE

The project gave hands-on experience with the full deep learning research lifecycle, from literature review and dataset selection through multimodal architecture design, comparative ablation, and rigorous evaluation. Three lessons stood out. First, fair comparison requires identical output heads; only the encoder should change between conditions. Otherwise the comparison conflates encoder quality with prediction-layer capacity. Second, pretraining is the single largest source of improvement on small datasets, well above any architectural sophistication added on top. Third, multimodal gains depend on whether modalities provide genuinely complementary information. When text describes features already obvious in the image (the carbohydrate result), adding text can hurt rather than help. The carbohydrate finding was the most instructive part of the project: a clean negative finding inside an otherwise positive overall comparison.

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